

Technical Datasheet — Grid Energy Storage Battery Cell

Model: VBF-T2R1MIMO-DSH-01
Date: 2026-08-30

Key Specifications

Parameter	Value	Unit
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Cell chemistry	NMC811 / Graphite	—
Nominal capacity	3.28	Ah
Nominal voltage	3.61	V
Energy density (gravimetric)	378.69	Wh/kg
Energy density (volumetric)	~1750	Wh/L (est.)
Max continuous charge rate	4C	—
Max continuous discharge rate	4C	—
Charge voltage limit	4.2	V
Discharge voltage limit	2.5	V
Operating temperature (charge)	-20 to 45	°C
Operating temperature (discharge)	-20 to 60	°C
Cycle life (80% retention)	>500	cycles
Calendar life	>10	years (est.)

Rate Performance

Rate	Capacity (Ah)	Retention vs 1C
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1C	3.28	100%
-20°C 1C	3.28	99.99%
4C charge	3.28	No plating

Safety

Test	Result
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4C charge @45°C max temperature	344.95 K (71.8°C) — PASS (≤350 K)
Lithium plating at 4C	Not detected (min anode V = +0.0006 V)

Durability

Condition	Metric	Value
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100 cycles @1C	SEI thickness	205.9 nm (≤500 nm)
500 cycles @1C	SEI thickness	518.1 nm (≤550 nm)